SYSTEM STATUS: ACTIVE | PILOT DEMO

N-SHIELD

AI-Driven Urban Safety Intelligence & Command Center

Turning fragmented city safety data into real-time actionable intelligence.

PILOT GEOGRAPHY

Nagpur, Maharashtra (6 Sectors)

CORE TECH STACK

Next.js 16 + Mapbox 3D + Deck.gl

TARGET STAKEHOLDERS

Police 112, Municipalities & Citizens

THE PROBLEM

Fragmented Safety Infrastructure & Delayed Emergency Response

Disconnected Data Silos

4+ Siloed Systems

Police FIR records, municipal CCTV feeds, citizen emergency calls (112), and streetlight lighting indices operate in isolation.

Reactive Patrol Allocation

Static Deployment

Police PCR vans and patrol units are assigned statically based on traditional beat boundaries rather than real-time dynamic crime density.

Unmonitored Surveillance Blind Spots

~40% Unmonitored

Authorities lack spatial visibility into CCTV coverage gaps across high-risk corridors, leading to guesswork in camera placement.

Uninformed Citizen Navigation

Zero Safety Factors

Conventional GPS navigation applications optimize purely for distance and travel time, routing pedestrians through unlit and unsafe alleys.

THE CORE IDEA

A Unified Intelligence Layer Over City Infrastructure

INCOMING CITY DATA FEEDS

- Crime Incident Reports
- CCTV Camera Nodes & Status
- Police Patrol GPS Telemetry
- Streetlight Lighting Indices
- Mass Gathering Metadata

N-SHIELD INTELLIGENCE ENGINE

[Algorithmic Risk Heatmapping (0-100)]

[Explainable AI (XAI) Directives]

[CCTV Gap & Blind Spot Analytics]

[Patrol Force Reallocation Engine]

[Predictive Event Impact Simulator]

[AI Safe Night Route Evaluator]

ACTIONABLE COMMAND OUTPUTS

3D Command Center Dashboard

1-Click Patrol Reallocation Directives

Citizen Safe Route & Night Mode

1-Tap Multi-Department SOS Dispatch

SYSTEM WORKFLOW

End-to-End Operational System Workflow

01

Data Ingestion

Ingests spatial incidents, CCTV status, patrol locations, lighting indices, and mass event data across Nagpur.

STATUS: ACTIVE

02

AI & Risk Computation

Executes multi-factor risk engine (R_{zone}), CCTV blind-spot clustering, and XAI directive generation.

STATUS: COMPUTING

03

Command Center GIS

Visualizes GPU-accelerated risk heatmaps, interactive markers, force deficit meters, and event scenarios.

STATUS: VISUALIZING

04

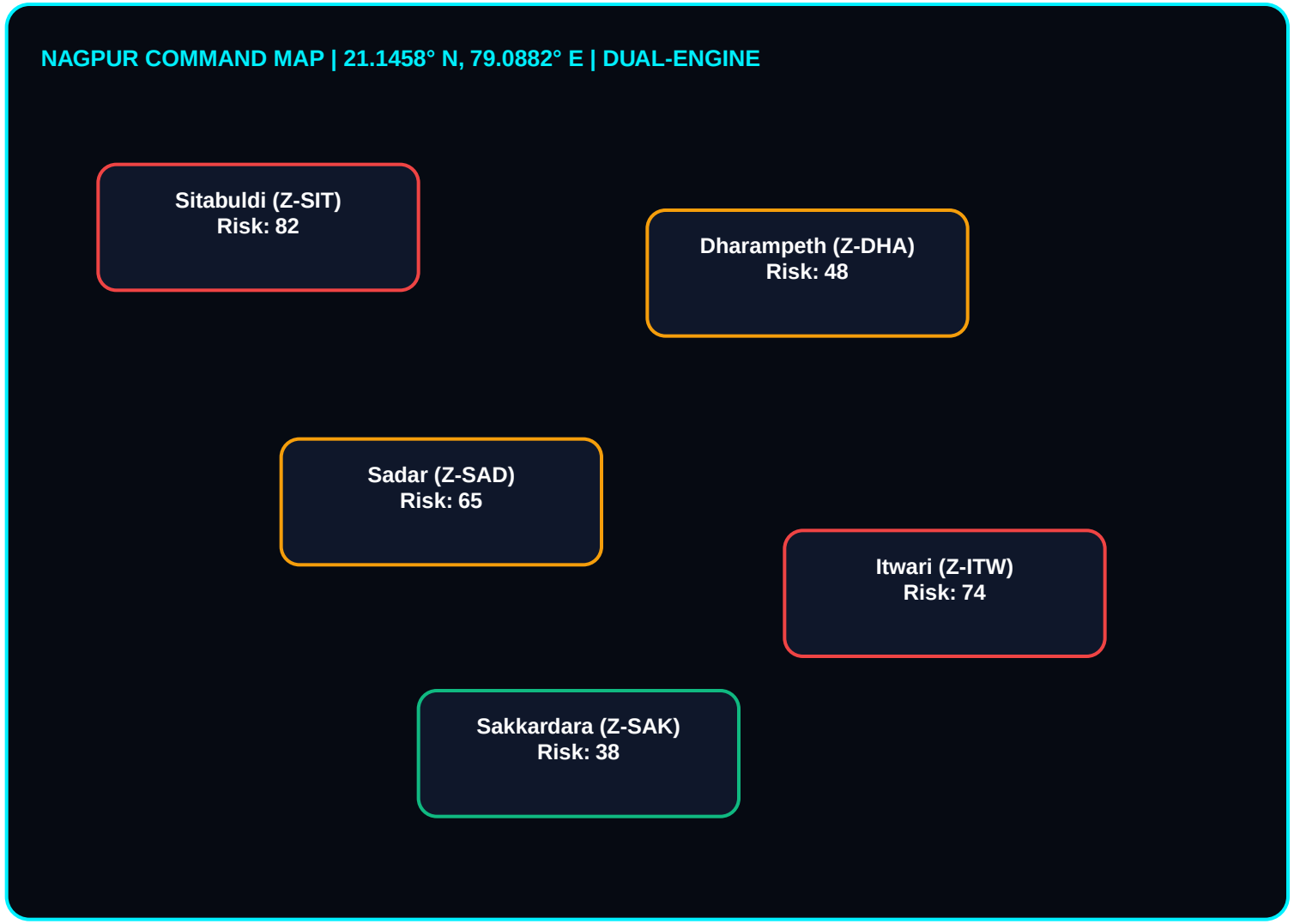
Targeted Dispatch

Outputs 1-click patrol reallocation to police, targeted CCTV expansion plans, and citizen route guidance.

STATUS: DISPATCHING

LIVE GIS COMMAND MAP

Interactive Command Map & 3D Digital Twin (Nagpur Pilot)



IMPLEMENTED GIS CAPABILITIES

- **Dual Mapping Engine**

  Google Maps vector API + Mapbox 3D extruded building basemap.

- **Deck.gl GPU Heatmaps**

  Hardware-accelerated real-time crime concentration layers.

- **Spatial Filter System**

  Category, severity, time period, and status incident filters.

- **Infrastructure Overlays**

  Toggle pins for CCTV, PCR vans, hospitals, and police stations.

STATUS: FULLY IMPLEMENTED IN REPOSITORY

AI RISK INTELLIGENCE

Algorithmic Risk Scoring (R_zone) & Explainable AI (XAI)

IMPLEMENTED RISK SCORING FORMULA

```
R_zone = Min(100, Rou
F_time * 0.27 + D_cctv * 0.2
```

- Incident Load (L_incident - 38%): Weighted by severity
- Time-of-Day Factor (F_time - 27%): Peak at 22:00-02:00 AM
- CCTV Deficit (D_cctv - 22%): Spatial unmonitored %
- Lighting Deficit (D_lighting - 13%): Night operational index

EXPLAINABLE AI (XAI) TRANSPARENCY

Deconstructs risk scores into transparent factor bars & tactical directives.

SECTOR: Sitabuldi (Z-SIT) | RISK SCORE: 82/100 (CRITICAL)

Historical Incident Load Contribution: 85%

CCTV Coverage Deficit Contribution: 70%

Night Vulnerability Index Contribution: 90%

AI TACTICAL DIRECTIVE:

"Deploy 2 Quick Response Bikes and establish temporary mobile CCTV unit at Sitabuldi station exit during 20:00 - 01:00 AM."

CCTV INTELLIGENCE

Surveillance Gap Analytics & Critical Blind Spot Detection

SPATIAL COVERAGE & BLIND SPOT ALGORITHMS

- Spatial Coverage Ratio = (Operational CCTV / Total Installed) * 100
- Critical Blind Spot: Incident Density > Threshold AND CCTV Coverage < 45%
- Hotspot Velocity (9Eö7 ime): ((N_current - N_prev) / N_prev) * 100 >= 15%
- Data-Driven Planning: Prioritizes municipal CCTV expansion algorithmically.

NAGPUR SECTOR SURVEILLANCE STATUS

Sitabuldi (Z-SIT)	42/60 (70%)	**Moderate Gap**
Dharampeth (Z-DHA)	55/65 (84%)	**Optimal**
Sadar (Z-SAD)	30/50 (60%)	**Moderate Gap**
Itwari (Z-ITW)	18/50 (36%)	**CRITICAL BLIND SPOT**
Sakkardara (Z-SAK)	22/45 (48%)	**High Gap**
Automotive Sq (Z-AUT)	12/40 (30%)	**CRITICAL BLIND SPOT**

RESOURCE & EMERGENCY

Algorithmic Patrol Force Allocation & Emergency SOS

POLICE PATROL DEPLOYMENT ENGINE

```
Req Patrols = Max(2, Round(P
```

• Deficit Analysis: Required Patrols - Active Patrols per sector.

• 1-Click Reallocation: Shifts units from low-risk (R < 35) to deficit sectors.

• Managed Units: PCR Vans, Quick Response Bikes, Interceptor Vehicles.

CITIZEN EMERGENCY SOS DISPATCH SYSTEM

Police Emergency (112)

Instant GPS broadcast to nearest PCR Van.

Medical Ambulance (108)

Direct route guidance to trauma hospitals.

Fire & Rescue (101)

Hazard alert & station dispatch.

Women Safety Helpline (1091)

Priority interceptor unit tracking.

AI SAFE NAVIGATION

Risk-Aware Safe Night Navigation Engine

ROUTE A — STANDARD FASTEST

- Estimated Travel Time: 12 Mins
- Distance: 4.2 km
- Risk Score: CRITICAL RISK (78/100)
- Street Lighting: 20% Lighting (Dark Alleys)
- Surveillance & Patrols: Low CCTV Coverage (15%)

NOT RECOMMENDED AT NIGHT

ROUTE B — N-SHIELD SAFE ROUTE

- Estimated Travel Time: 15 Mins (+3 min)
- Distance: 4.8 km
- Risk Score: LOW RISK (22/100)
- Street Lighting: 90% Lighting (Main Arterials)
- Surveillance & Patrols: High CCTV & PCR Presence

RECOMMENDED SAFE ROUTE

NIGHT SAFETY INTELLIGENCE

Dark-Spot Heatmapping & Web Audio Acoustic Alarms

NIGHT-TIME THREAT MONITORING

• Dynamic Night Lighting Deficit: Streetlight efficiency drops in calculations at 21:00 - 04:00 AM.

• Dark Spot Heatmapping: Maps unlit streets & nighttime threat zones across Nagpur.

• Synthesized Web Audio Alarm: Uses Web Audio API (audioService.ts) to sound acoustic warnings.

• Crowdsourced Citizen Reporting: Citizens report unlit streets to municipal electrical boards.

LIVE NIGHT SAFETY MODE CONTROL

MODE: NIGHT SAFETY ACTIVE
ACOUSTIC ALARM: ENABLED (Web Audio API)

UNLIT CORRIDOR ALERT: AUTOMOTIVE SQ (Z-AUT)
LIGHTING INDEX: 12% | NIGHT THREAT SCORE: 88/100

STATUS: FULLY IMPLEMENTED IN REPOSITORY

EVENT SIMULATOR

Mass Gathering & Public Event Risk Modeling Engine

PREDICTIVE EVENT IMPACT ENGINE

• Select Event Scenario: VCA Stadium Cricket Match, Ganesh Visarjan Procession, Political Rally.

• Dynamic Variable Adjustments: Injects crowd surge density & traffic congestion multipliers.

• Sector Risk Recalculation: Recalculates target sector risk score (e.g. Dharampeth risk 48 -> 85).

• Automated Force Shift Generation: Calculates additional police PCR vans required for pre-event staging.

SIMULATION CASE STUDY: VCA STADIUM MATCH

Baseline Sector Risk

48 / 100 (Medium)

Estimated Footfall Surge

+45,000 Spectators

Simulated Post-Event Risk

86 / 100 (Critical)

Auto-Recommended Reinforcements

+4 PCR Vans, +6 QR Bikes

DIFFERENTIATION MATRIX

Traditional Fragmented Systems vs. N-SHIELD Unified Command

SAFETY DIMENSION	TRADITIONAL APPROACH	N-SHIELD PLATFORM
Data Visibility	Disconnected paper FIRs & isolated digital files	**Unified 3D Digital Twin Command Dashboard**
Risk Assessment	Static historical annual crime statistics	**Dynamic multi-factor risk engine (R_zone)**
CCTV Infrastructure	Monitored individually without spatial context	**Coverage gap & critical blind spot analytics**
Citizen Navigation	Optimized purely for distance & travel time	**Multi-variable safety & lighting aware routing**
Patrol Allocation	Static beat patrols based on traditional boundaries	**Algorithmic deficit-based auto-reallocation**
Event Management	Manual security planning & reactive deployment	**Real-time predictive event impact simulation**

SYSTEM ARCHITECTURE

Production Enterprise Technology Stack & Architecture

IMPLEMENTED TECH STACK

• Core Web Framework

Next.js 16 (App Router), React 19, TypeScript

• Styling & UI

Tailwind CSS v4, Lucide Icons, Recharts, Framer Motion

• GIS & Digital Twin

Google Maps API (@vis.gl), Mapbox GL JS, Deck.gl

• Audio Engine

Synthesized Web Audio API Alarm (audioService.ts)

• REST API Endpoints

Next.js APIs: /api/incidents, /api/cctv, /api/zones

DATA & ENGINE ARCHITECTURE

Presentation Layer

Next.js 16 React Client + Google Maps / Mapbox Canvas

API Gateway

REST APIs (/api/*) + Real-time RSS News Feed Service

Intelligence Layer

RiskEngine + BlindSpotIntelligence + DeploymentEngine

City Data Infrastructure

Nagpur Sector Coordinates, CCTV Nodes & Incident Schema

REAL-WORLD IMPACT

Multi-Stakeholder Value & City-Wide Expansion Path

Law Enforcement / Police 112

- Real-time 3D GIS situational awareness
- Deficit-based force reallocation
- XAI transparent tactical directives

Smart City Authorities

- Data-driven CCTV installation planning
- Streetlight maintenance prioritization
- Mass event safety risk management

Citizens & Tourists

- Risk-aware safe night navigation
- 1-Tap multi-department SOS dispatch
- Crowdsourced dark-spot reporting

Emergency Medical & Fire

- Sub-second location identification
- Direct trauma route guidance
- Multi-agency coordination

ROADMAP & VISION

Phased Implementation Roadmap & Strategic Vision

PHASE 1 (CURRENT)

Interactive Prototype & 6-Sector Command Center Model

PHASE 2 (MONTHS 1-3)

Integration with Nagpur Police CCTNS FIR Database

PHASE 3 (MONTHS 3-6)

IoT Smart Streetlight & Acoustic Sensor Ingestion

PHASE 4 (MONTHS 6-12)

Spatio-Temporal GNN Crime Forecasting Models

STRATEGIC CLOSING VISION

"N-SHIELD does not replace law enforcement; it empowers city decision-makers with a unified intelligence layer to transform urban safety from reactive incident management into proactive, data-driven protection."